

Lab 2: Connect to Cloud (40 points)

Handout: 5 Oct, Hand-in: 13 Oct (Tuesday)

Overview

In this lab, your group will work together to connect your MakePython ESP32 to the AWS cloud.

Task (12 points): Connect temperature sensor DS18B20 and MakePython ESP32, then make it works properly.

Requirements:

1. Show the app source code is successfully uploaded onto the board with screenshot.
2. Show that the temperature sensor is working properly with a photo of your boards screen. The screen should show the temperature.



Task (20 points): Set up the MakePython ESP32 and Amazon AWS to forge a edge-cloud IoT application.

Hints: Properly set up your SUTD-WIFI.

Requirements:

1. Show that the temperature sensor is working properly with a photo of your boards screen. The screen should show the temperature periodically.



2. Show that the AWS is able to receive the temperature data with screenshots.

▼ esp32/pub

```
{  
  "device": "ESP32-temp",  
  "temperature_celsius": 27.6875,  
  "temperature_fahrenheit": 81.8375,  
  "timestamp": 315118  
}
```

Answer following questions (8 points):

1. Please briefly describe the MQTT communication process. You need to use source code for explanation.

What to hand-in?

Write a brief **lab report** for the tasks above:

- **Deliverables:** Provide evidence of the completion of all tasks and fulfilment of the requirements.
- **Reflection:** Reflect on your experience or problems during the setup process.
- **Acknowledgement:** Acknowledge any students, whether in or outside your group, who assisted you during the lab.

Summary of Online Resources

1. TA's example of connecting to AWS by MQTT, using SUTD-Wifi. (ESP-IDF: Only for MakePython, **please use the 'lab1_2_1' branch for task 1, 'lab1_2_2' for task 2**):<https://github.com/TONYSKYZENG/CCIOTLabs.git>
Arduino: https://github.com/nan2li/SUTD_CCIOT_AWS-IOT-core_ESP32_PCAP-wifi/tree/main)
2. ESP's example of connecting to AWS by MQTT, using WPA(2) WIFI.
https://github.com/espressif/esp-aws-iot/tree/master/examples/mqtt/tls_mutual_auth
3. ESP's example of connecting to enterprise wifi like SUTD-Wifi:
https://github.com/espressif/esp-idf/tree/release/v4.3/examples/wifi/wpa2_enterprise